

KOTKA RANKKI, Finland

WMO: 029760

Lat: 60.375N

Lon: 26.959E

Elev: 43

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.7	0.6	-8.8	3.4	-4.6	-3.3	4.6	0.8	32.8	31.5	29.1	29.6	8.7	10	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	76.2	68.2	73.6	66.7	71.1	64.9	70.0	74.3	67.8	72.6	65.8	70.1	8.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	104.1	73.1	65.9	95.6	71.0	63.8	88.8	68.9	33.9	74.2	32.2	72.9	30.6	70.3	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	25.3	22.6	-5.8	79.7	8.5	4.2	-11.9	82.7	-16.9	85.1	-21.7	87.4	-27.9	90.5		
(5)			-6.3	70.8	8.4	2.7	-12.3	72.7	-17.3	74.3	-22.0	75.8	-28.1	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	24.1	22.2	27.7	37.1	47.8	57.0	64.1	62.9	54.8	44.4	36.2	30.0		
	DBStd	15.93	10.04	10.87	7.32	5.07	6.09	5.22	5.11	4.55	5.06	5.98	6.62	8.54		
	HDD50	4019	802	779	692	389	115	4	0	16	188	414	620			
	HDD65	8313	1267	1199	1157	838	535	245	79	307	640	864	1085			
	CDD50	1273	0	0	0	1	46	215	438	399	160	14	0	0		
	CDD65	88	0	0	0	0	0	6	52	30	0	0	0	0		
	CDH74	188	0	0	0	0	0	10	134	44	0	0	0	0		
	CDH80	10	0	0	0	0	0	1	9	0	0	0	0	0		
	WSAvg	11.1	12.4	11.8	11.2	9.9	9.6	9.6	8.9	9.6	10.8	12.5	13.4	13.8		
	PrecAvg	22.8	1.7	1.7	1.3	1.2	1.4	2.2	2.0	2.4	2.1	2.3	2.4	1.8		
(15) Precipitation	PrecMax	27.8	3.3	3.9	2.5	2.5	3.3	5.1	4.0	5.0	4.1	4.4	5.2	3.2		
	PrecMin	16.3	0.3	0.1	0.2	0.3	0.3	0.2	0.5	0.1	0.5	0.7	0.3	0.4		
	PrecStd	3.0	0.7	1.1	0.6	0.7	0.9	1.2	1.1	1.3	1.1	0.9	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	37.7	42.2	55.6	70.1	75.1	81.4	78.0	67.6	57.5	49.2	44.8		
		MCWB	38.9	35.2	37.4	48.2	58.0	65.1	70.9	68.8	61.3	54.8	48.1	43.6		
	2%	DB	37.9	35.7	39.0	49.6	64.1	70.8	77.7	74.9	64.9	55.5	47.8	43.1		
		MCWB	37.1	34.6	35.7	43.7	55.7	62.8	69.0	67.4	60.1	53.4	46.7	42.2		
	5%	DB	36.3	34.7	36.9	46.7	60.3	67.8	75.0	72.5	63.2	53.9	46.4	41.5		
		MCWB	35.6	33.9	34.6	41.8	54.0	61.0	67.9	65.7	59.2	52.1	45.2	40.6		
	10%	DB	34.8	33.8	35.3	44.1	56.6	65.2	72.6	70.0	61.7	52.3	44.8	39.6		
		MCWB	34.0	33.2	33.8	40.3	51.1	59.4	66.6	64.2	58.3	50.5	43.8	38.6		
	0.4%	WB	39.2	35.9	38.4	48.3	60.0	67.7	73.7	71.4	63.2	55.7	48.5	44.1		
		MCDB	39.4	36.8	40.9	54.1	65.3	73.5	78.2	75.7	65.6	56.9	48.8	44.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.4	35.0	36.4	45.0	57.1	64.6	71.2	68.9	61.5	53.8	46.8	42.4		
		MCDB	37.8	35.4	38.0	48.8	62.3	69.2	75.3	73.5	63.7	55.2	47.5	42.9		
	5%	WB	35.7	34.1	35.1	42.6	54.6	62.3	69.2	66.9	60.2	52.5	45.5	40.6		
		MCDB	36.2	34.5	36.3	45.6	59.6	66.4	73.7	71.3	62.3	53.8	46.3	41.2		
	10%	WB	34.2	33.3	34.1	40.8	51.8	60.0	67.1	65.0	58.9	50.8	43.9	38.9		
		MCDB	34.7	33.7	35.0	43.5	56.2	64.2	71.8	68.8	61.2	52.1	44.6	39.6		
	5% DB	MDBR	7.0	7.8	9.1	8.7	9.9	9.4	9.4	8.6	7.2	5.5	5.1	5.9		
		MCDBR	4.8	5.3	8.2	12.0	14.3	13.0	12.0	10.9	8.4	5.5	5.1	4.5		
		MCWBR	5.0	5.1	5.9	7.3	7.6	6.7	5.9	5.6	5.6	5.0	5.5	4.7		
		MCWBR	4.9	5.1	6.8	10.8	12.8	11.1	10.5	9.2	7.0	5.2	4.8	4.3		
(35) Clear-Sky Solar Irradiance	taub		0.278	0.311	0.333	0.371	0.369	0.365	0.387	0.380	0.353	0.334	0.305	0.382		
		taud	2.139	2.167	2.215	2.285	2.404	2.458	2.417	2.433	2.489	2.388	2.207	2.596		
	Ebn at Noon		148	211	248	261	272	275	265	257	245	206	136	103		
		Edh at Noon	13	23	30	35	33	32	33	30	24	19	13	9		
(44) All-Sky Solar Radiation	RadAvg		80	273	745	1227	1682	1751	1751	1335	793	328	88	43		
	RadStd		17	58	93	133	181	116	157	137	93	49	15	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (51 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	-299	-317	N/A		

Nomenclature: See separate page